MATH317 · NUMERICAL METHODS

Homework 1 Practice

Errors · Taylor · Horner

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01 Questions

Answer each question in the space below it. The full solution is in a separate file.

Q1 RELATIVE ERROR

A real number x is represented approximately by 0.6032, and the relative error is at most 0.1%. What is x?

Q2 ROUNDING

What is the relative error involved in rounding 4.9997 to 5.000?

Q3 MACLAURIN

Derive the Maclaurin series of $(1 + x)^n$. Write it for $n = 2$, $n = 3$ and $n = \frac{1}{2}$. Then use it to compute $\sqrt{1.0001}$ correct to 15 decimal places.

Q4 NUMBER OF TERMS

How many terms of the series for $\ln(1 + x)$ are needed to compute $\ln 1.1$ with an error less than $\frac{1}{2} \times 10^{-8}$?

Q5 TAYLOR

Write the Taylor expansion of $f(x) = x^3 - 2x^2 + 4x - 1$ in the form $f(2 + h)$.

Q6 COMPLETE HORNER

Use the complete Horner algorithm to find the Taylor expansion of $p(x) = 3x^5 - 2x^4 + 15x^3 + 13x^2 - 12x - 5$ about $r = 2$.

Q7 COMPLETE HORNER

Find the Taylor series of $f(x) = x^4 - 4x^3 + 7x^2 - 5x - 2$ about $c = 3$.